TEXORA · LECTURE SHEET

Thermal Properties

Heat and the fibre: thermal conductivity and why fabrics insulate by trapping air, specific heat, the glass transition and melting that organise all heat processes, thermal contraction, heat setting — and the fire behaviour that safety standards are written around.

TE-3209 · TEXTILE PHYSICS · TOPIC 7 · LEVEL 3 TERM 2

1. Thermal conductivity — air does the insulating

Measured along the fibre, conductivities (typical reported values, W/m·K) rank: cotton ≈ 0.46, viscose ≈ 0.29, nylon ≈ 0.24–0.28, wool ≈ 0.19, polyester ≈ 0.14 — but **still air is 0.026**, an order below them all. A fabric is mostly air by volume, so its insulation is governed by how much still air it traps and how well it stops that air moving: thickness of trapped air rules warmth, which is why lofty wool and hollow-fibre fills insulate, why windproof shells protect insulation, and why compressing a duvet wrecks it. The ranking of the fibres themselves matters at the contact scale — the cool touch of linen/cotton (high conductivity pulls heat from skin quickly) versus the warm touch of wool and acrylic.

2. Transitions — Tg and Tm run the process calendar

Heat a fibre and its structure passes landmarks. At the **glass transition** Tg, amorphous chain segments gain mobility: modulus falls, dyeing becomes possible at practical rates (disperse dyes enter PET only above its Tg ≈ 75 °C in water-plasticised form), and shapes can be permanently re-set. At the **melting point** Tm (synthetics with crystallites), order collapses: PET and nylon 6.6 melt ≈ 255–260 °C, nylon 6 ≈ 215–220 °C. Natural and regenerated fibres have no melt — cellulose and protein degrade and char instead (cotton yellows with prolonged heat above ≈ 150 °C and decomposes near 240 °C+; wool degrades from ≈ 130 °C prolonged exposure).

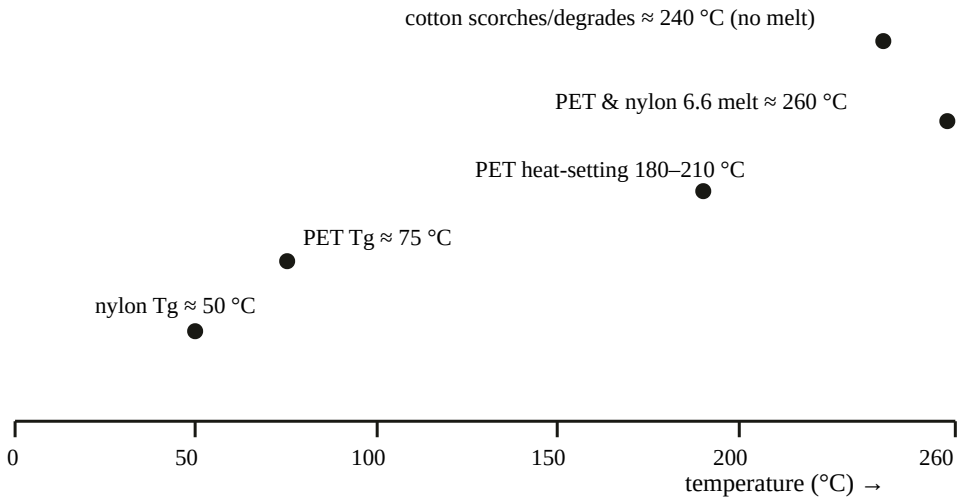


Figure 1 — Thermal landmarks (axis 0–260 °C): glass transitions open dyeing and setting; melts end the synthetics' road; cellulosics degrade without melting.

Heat setting exploits the transitions: hold a thermoplastic fabric at 180–210 °C (PET) under controlled dimensions, chains rearrange into the new configuration, and on cooling the shape is locked — permanent pleats, dimensional stability, stenter settings. Set too cool and the memory is weak; set in the dye bath temperature's reach and later dyeing unsettles it — the process calendar of a polyester plant is written in Tg and setting temperature.

3. Thermal expansion, contraction and specific heat

Oriented fibres **contract on heating** along the axis (negative thermal expansion): heating mobilises chains whose orientation is a stretched, low-entropy state, and they recoil — the shrinkage of filaments in hot water or the stenter, harnessed deliberately in high-shrinkage yarns for bulked fabrics, and feared as garment shrinkage. Specific heats sit near water's third: cotton ≈ 1.2–1.3, wool ≈ 1.3, PET ≈ 1.0–1.3 kJ/kg·K (rising with temperature and, for hydrophilic fibres, with regain, since the sorbed water brings its own 4.18). These numbers feed every dryer and thermo-fixation energy balance in the mill.

4. Fire behaviour

Burning character divides the fibres sharply. The **limiting oxygen index** (LOI, % O_2 needed to sustain burning) ranks: cotton 18, viscose 19, nylon 20–21, polyester 20–21, wool 25, modacrylic ≈ 28–30, aramids 28–30+ (air is 21% O_2, so fibres below 21 burn freely in air). Behaviour matters as much as the number: thermoplastics **melt and drip** (shrinking away from small flames — sometimes self-protecting, sometimes spreading burns and causing melt-contact injuries); cotton burns steadily without melting; wool chars, self-extinguishes off the flame and is the traditional protective fibre (blankets, aircraft seats); FR finishes and inherently FR fibres engineer the rest. Ironing temperatures in care

labels are the domestic echo of the same ladder — cotton 200 °C, wool/silk/polyester ≈ 150 °C, nylon and acrylic ≈ 135 °C maximum soleplate.

5. Key points

- Fabrics insulate by trapped still air (k = 0.026 W/m·K); fibre conductivity ranking (cotton 0.46 → polyester 0.14) mainly sets the touch feel.
- Tg unlocks dyeing and setting (PET ≈ 75 °C); Tm ends thermoplastics (≈ 260 °C); naturals degrade instead of melting.
- Heat setting at 180–210 °C writes permanent shape memory into PET; later hot processes must respect the set temperature.
- Oriented fibres contract on heating (entropy recoil) — shrinkage as nuisance and as texturing tool.
- LOI: cotton 18 and wool 25 bracket the apparel range; melt-drip vs char behaviour decides injury patterns and FR strategy.

6. Practice

Answer in your own words; brief model answers follow.

1. Two identical-thickness fleeces, one wool-rich and lofty, one dense cotton, are compared for warmth. Predict the winner and give the mechanism — and the condition (wind) that could reverse the practical result.
2. Why must disperse dyeing of polyester run at 130 °C (HTHP) or use carriers, when cotton dyes at the boil with direct dyes?
3. A polyester fabric heat-set at 190 °C is later dyed at 130 °C and remains stable, but a fabric set at 120 °C distorts in the same dye bath. Explain via chain mobility.
4. Rank cotton, wool, polyester and nylon 6 by LOI and state which are “self-extinguishing in air” by the LOI > 21 criterion.
5. Estimate the energy to heat 500 kg of dry cotton fabric ($c_p \approx 1.3$ kJ/kg·K) from 30 °C to 150 °C in a stenter, ignoring moisture.
6. Why do ironing guides permit 200 °C for cotton but only ≈ 135 °C for nylon, although nylon’s melting point (≈ 215 °C for nylon 6) is far above 135 °C?

Answer key

1. The lofty wool fleece: it traps more still air per thickness (crimp holds the loft), and insulation is trapped-air thickness. In wind without a shell, air exchanges through the open structure and the denser cotton may feel warmer — windproofing protects the mechanism, not the fibre.

2. Disperse dyes are non-ionic guests that must diffuse into PET’s amorphous regions; below Tg the chains are glassy and diffusion is negligible. At 130 °C (or with a carrier plasticising the surface) segmental mobility opens the structure. Cotton’s open, water-swollen amorphous network admits large dye ions at 100 °C without needing a transition.
3. Setting at 190 °C rearranged chains into configurations stable up to that temperature; 130 °C cannot mobilise what 190 °C locked. The 120 °C set is weaker than the dye bath itself — the bath becomes the hotter “set”, and the fabric moves to the bath’s dimensions: distortion.
4. LOI: cotton 18 < viscose 19 (if included) < nylon 6 ≈ 20–21 ≈ polyester < wool 25. By the simple >21 rule only wool qualifies among the four (aramids and FR fibres exceed it properly); note LOI is a laboratory ranking, not a complete fire-safety verdict.
5. $Q = mc\Delta T = 500 \times 1.3 \times 120 = 78{,}000$ kJ = 78 MJ — before the (usually larger) moisture evaporation duty, which is why dryer energy audits start with water removal.
6. Because damage starts far below melting: nylon softens, glazes, yellows and loses strength at iron temperatures well under Tm; the safe soleplate is set by glazing/degradation and sticking risk, not by the melt. Cotton tolerates brief 200 °C contact because its degradation in seconds-scale exposure is negligible.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib